is weighted k means clustering of precipitation zones using weights of topographic and environmental variables derived from random forest an up-to-date approach?

Weighted K-means clustering combined with machine learning-derived variable weights, such as those from random forest models, represents a modern and effective approach for delineating precipitation zones. This method leverages the relative importance of topographic and environmental variables to improve clustering accuracy and spatial homogeneity, as demonstrated in agricultural and hydrological applications where weighted clustering outperforms traditional K-means (Ohana-Levi et al., 2019). Random forest models are commonly used to derive these weights due to their ability to capture nonlinear relationships and variable importance, which enhances the clustering process (Sreeparvathy & Srinivas, 2022; Tang et al., 2022). While direct studies specifically combining weighted K-means clustering of precipitation zones with random forest-derived weights are limited, related research shows that integrating machine learning with clustering improves precipitation forecasting and regionalization (Sreeparvathy & Srinivas, 2022; Tang et al., 2022; Zhang et al., 2016). Additionally, the use of cluster validity indices and spatial analysis further refines the identification of homogeneous precipitation regions, supporting the robustness of weighted clustering approaches (Bhatia et al., 2020; Zhang et al., 2016). Overall, this approach aligns well with current trends in hydrological and environmental modeling, making it an up-to-date and promising method.

These papers were sourced and synthesized using Consensus, an AI-powered search engine for research. Try it at https://consensus.app

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